Submission File: Code File (`ransomware_assessment.py`)

Author: Ritika Sonali Suryawanshi | Sandip University, Nashik | Project Topic #10

Below is the complete source code for the Enterprise Ransomware Assessment Tool and Risk Register Generator mapped to the NIST CSF 2.0 framework:

```
import json
import csv

# Enterprise Ransomware Assessment Tool & Risk Register Generator
# Mapped to NIST Cybersecurity Framework (CSF 2.0)
# Author: Ritika Sonali Suryawanshi | Sandip University, Nashik

RISK_REGISTER = [
    {
        "risk_id": "R-01",
        "scenario": "Spear phishing emails delivering ransomware loaders via malicious links.",
        "nist_function": "Protect (PR)",
        "nist_category": "PR.AT-01 (Awareness Training)",
        "likelihood": "High",
        "impact": "Critical",
        "risk_score": "High",
        "mitigation": "Mandatory phishing simulation, email filtering (SPF/DKIM/DMARC), EDR.",
        "residual_risk": "Low"
    },
    {
        "risk_id": "R-02",
        "scenario": "Lateral movement via exposed RDP and weak Active Directory accounts.",
        "nist_function": "Protect (PR)",
        "nist_category": "PR.AC-01 (Access Control)",
        "likelihood": "High",
        "impact": "High",
        "risk_score": "High",
        "mitigation": "Enforce Multi-Factor Authentication (MFA), disable public RDP, implement
LAPS.",
        "residual_risk": "Medium"
    },
    {
        "risk_id": "R-03",
        "scenario": "Exploitation of unpatched software vulnerabilities in edge VPN gateways.",
        "nist_function": "Identify (ID)",
        "nist_category": "ID.RA-01 (Risk Assessment)",
        "likelihood": "Medium",
        "impact": "Critical",
        "risk_score": "High",
        "mitigation": "Automated patch management, vulnerability scanning, threat intelligence.",
        "residual_risk": "Low"
    },
    {
        "risk_id": "R-04",
        "scenario": "Encryption of backup systems during a double-extortion ransomware attack.",
        "nist_function": "Recover (RC)",
        "nist_category": "RC.RP-01 (Recovery Plan)",
        "likelihood": "Medium",
        "impact": "Critical",
        "risk_score": "High",
        "mitigation": "Maintain offline/immutable backups (3-2-1 rule), test restoration drills.",
        "residual_risk": "Low"
    }
]

def generate_csv_report(output_filename="corporate_risk_register.csv"):
    fieldnames = list(RISK_REGISTER[0].keys())
    with open(output_filename, mode='w', newline='', encoding='utf-8') as f:
        writer = csv.DictWriter(f, fieldnames=fieldnames)
```